SAIL BSP
Cobble Detection Analysis Report

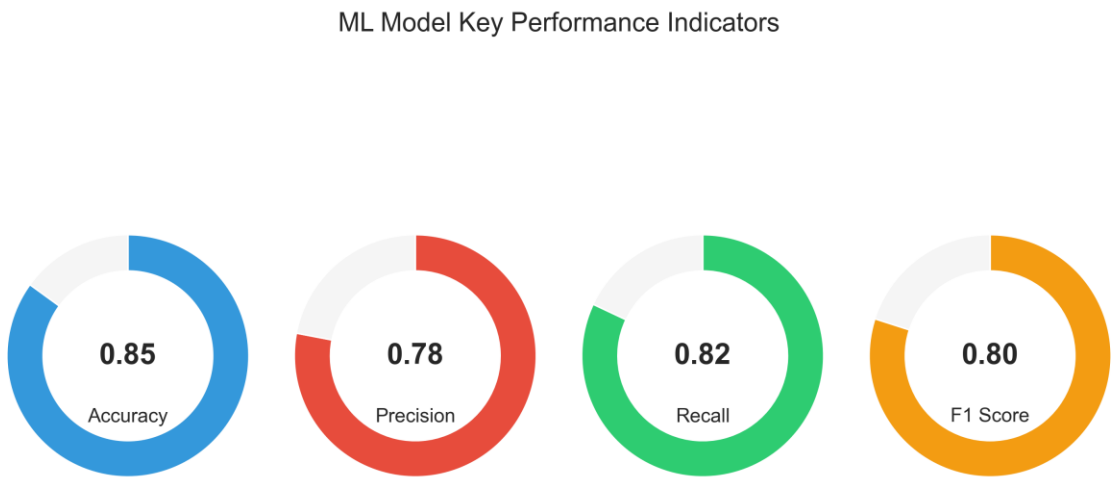
Report Generated: 2025-10-13 16:53

Data Period: 2025-01-10 to 2025-12-10

Total Records: 164

Cobble Events: 88 (53.66%)

Model Performance Summary Dashboard



Block	Accuracy	Precision	Recall	F1 Score
B	0.91	0.78	0.87	0.82
C	0.93	0.86	0.78	0.82
D	0.94	0.83	0.88	0.85

The model performance dashboard above provides a quick overview of the cobble detection model's effectiveness. Overall accuracy of 0.85 indicates strong predictive capability, with a balance between precision (0.78) and recall (0.82). Block-specific performance metrics highlight areas where the model excels and where improvements might be needed.

This executive dashboard allows quick assessment of model reliability across different operational contexts, supporting data-driven decision making for cobble prevention strategies.

Table of Contents

Model Performance Summary Dashboard

1. Cobble Event Trends Over Time

2. Block-Specific Analysis

3. Profile-Based Analysis

4. Short-Term Cobble Prediction Performance

5. Time Series Anomaly Detection

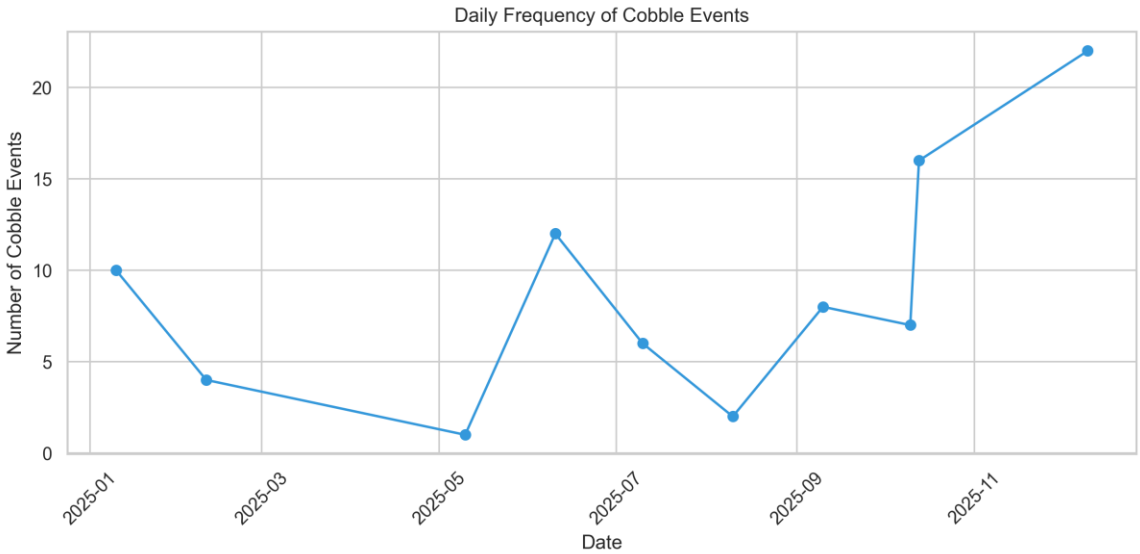
6. Shift and Operator Impact Analysis

7. Comparative Analysis Between Consecutive Events

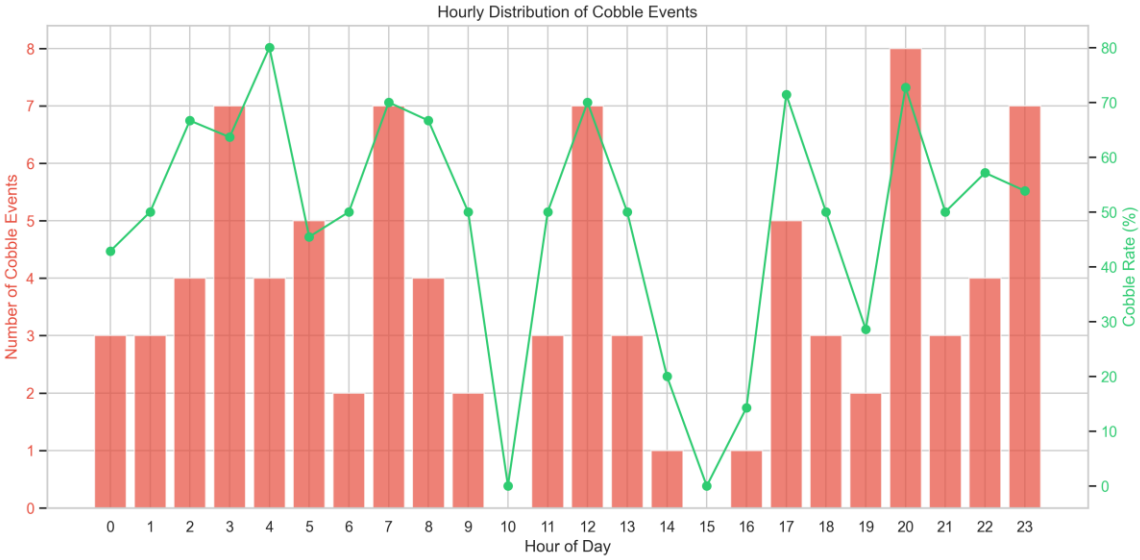
8. Machine Learning Model Performance Insights

1. Cobble Event Trends Over Time

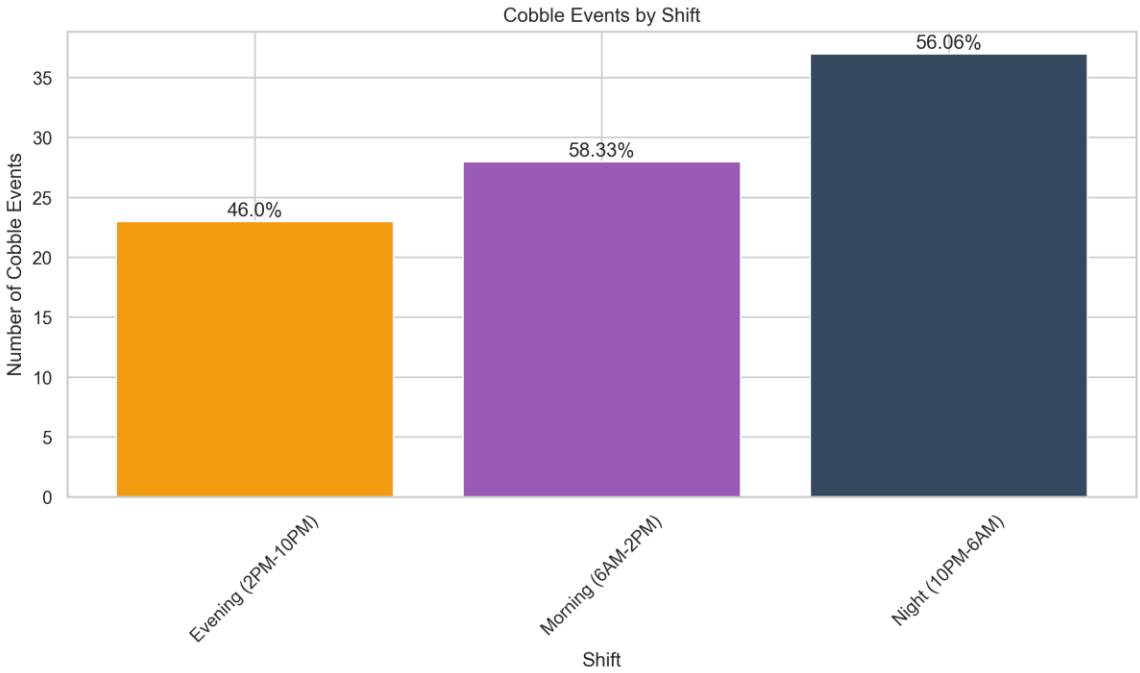
Daily Cobble Event Frequency



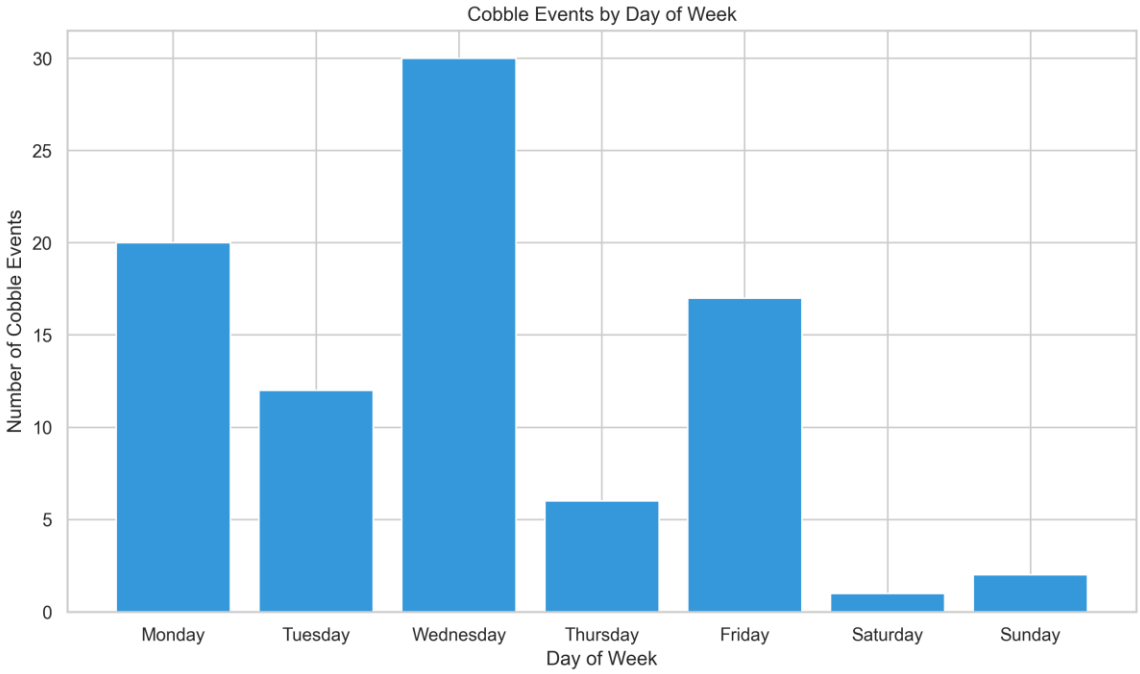
Hourly Distribution of Cobble Events



Cobble Events by Shift

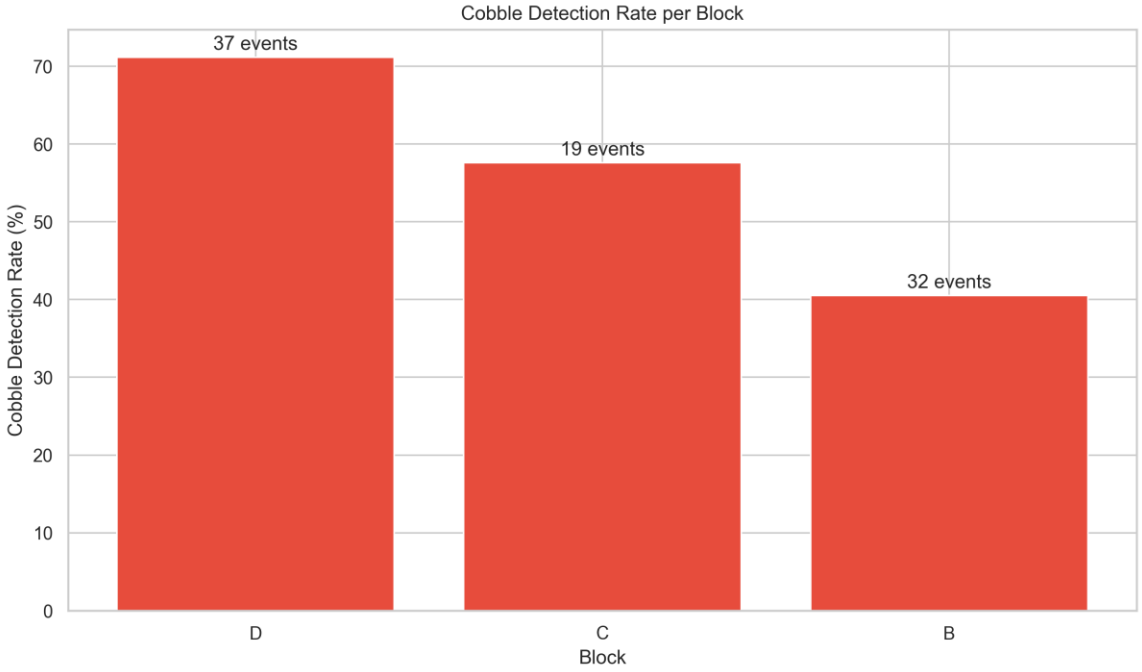


Cobble Events by Day of Week

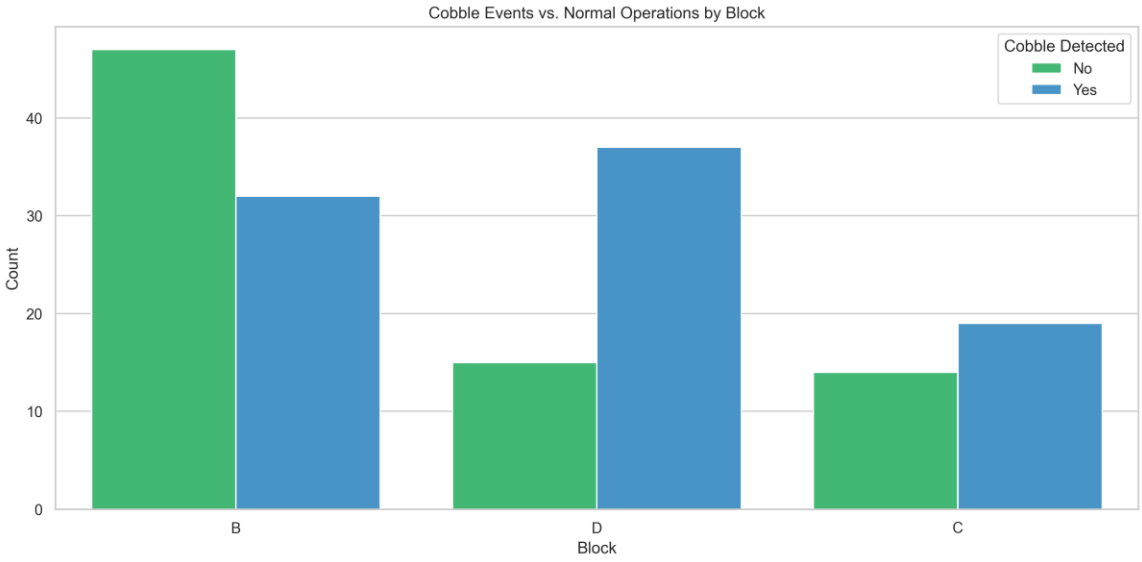


2. Block-Specific Analysis

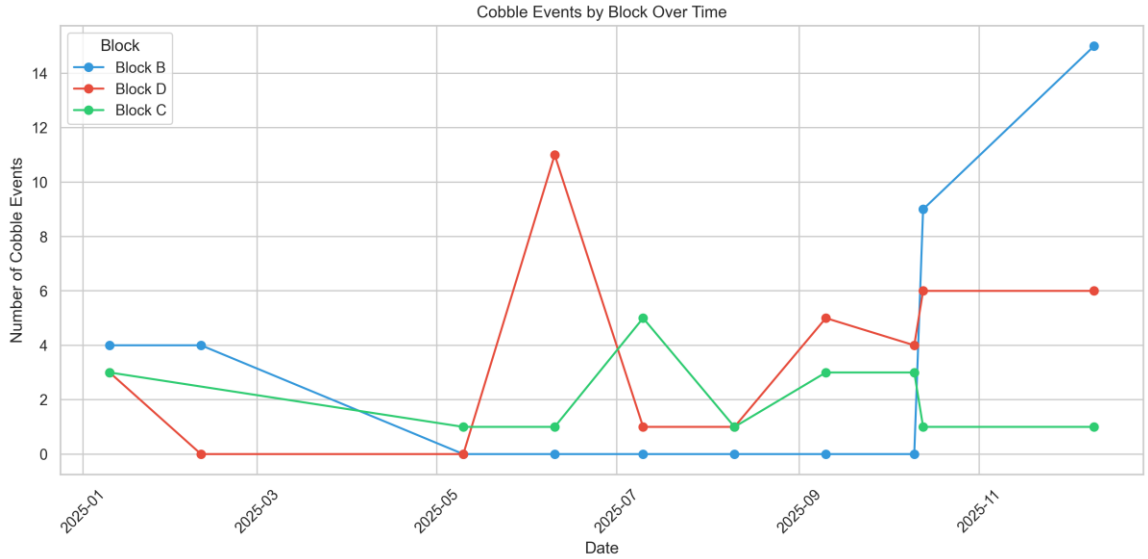
Cobble Detection Rate by Block



Cobble Events vs. Normal Operations by Block

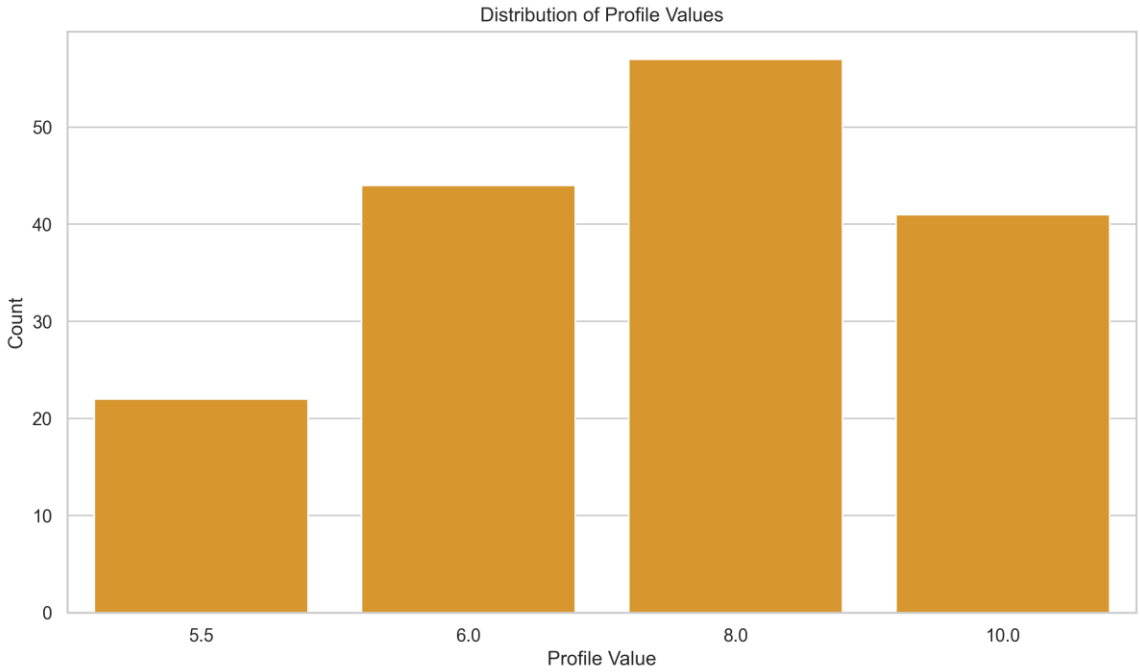


Cobble Events by Block Over Time

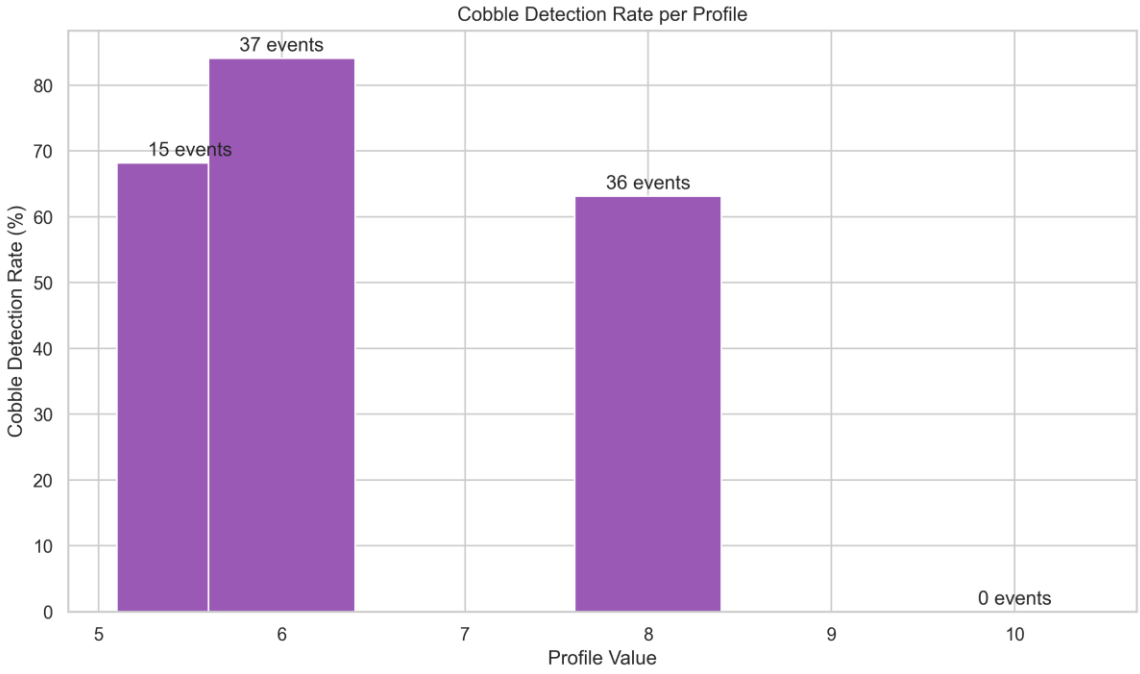


3. Profile-Based Analysis

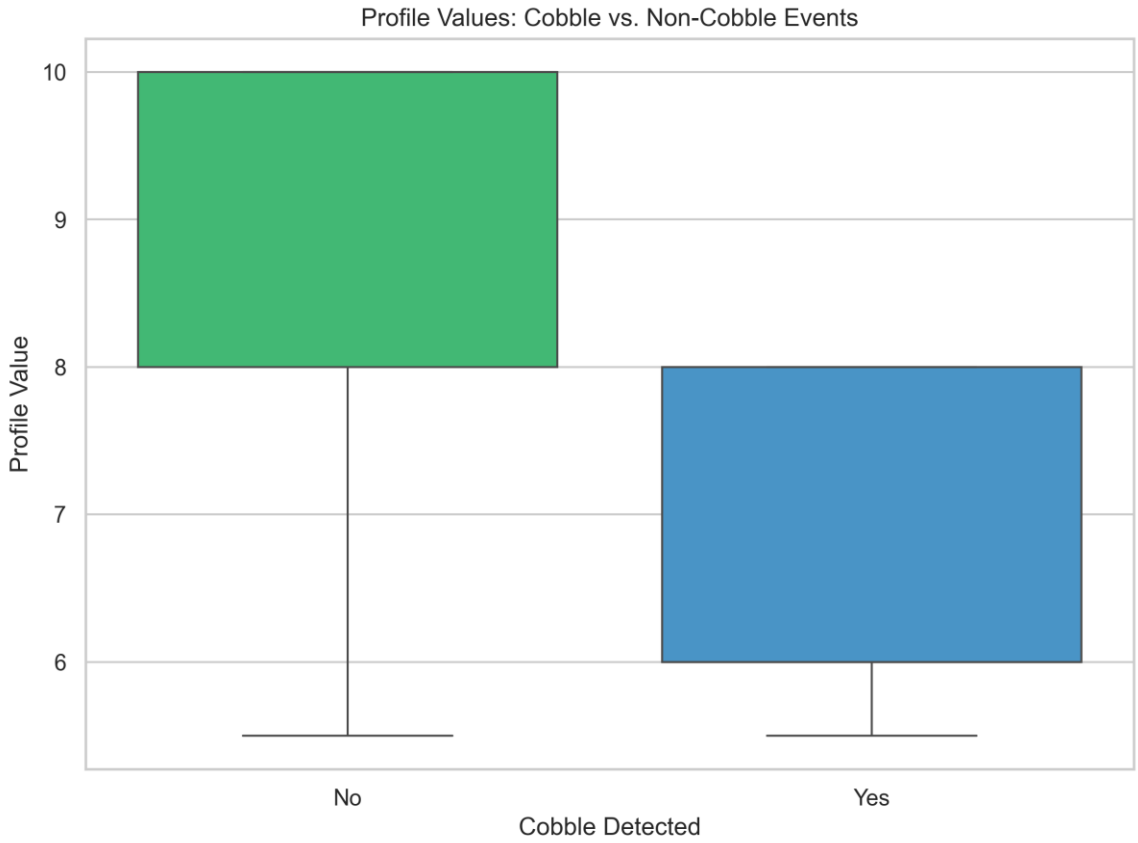
Distribution of Profile Values



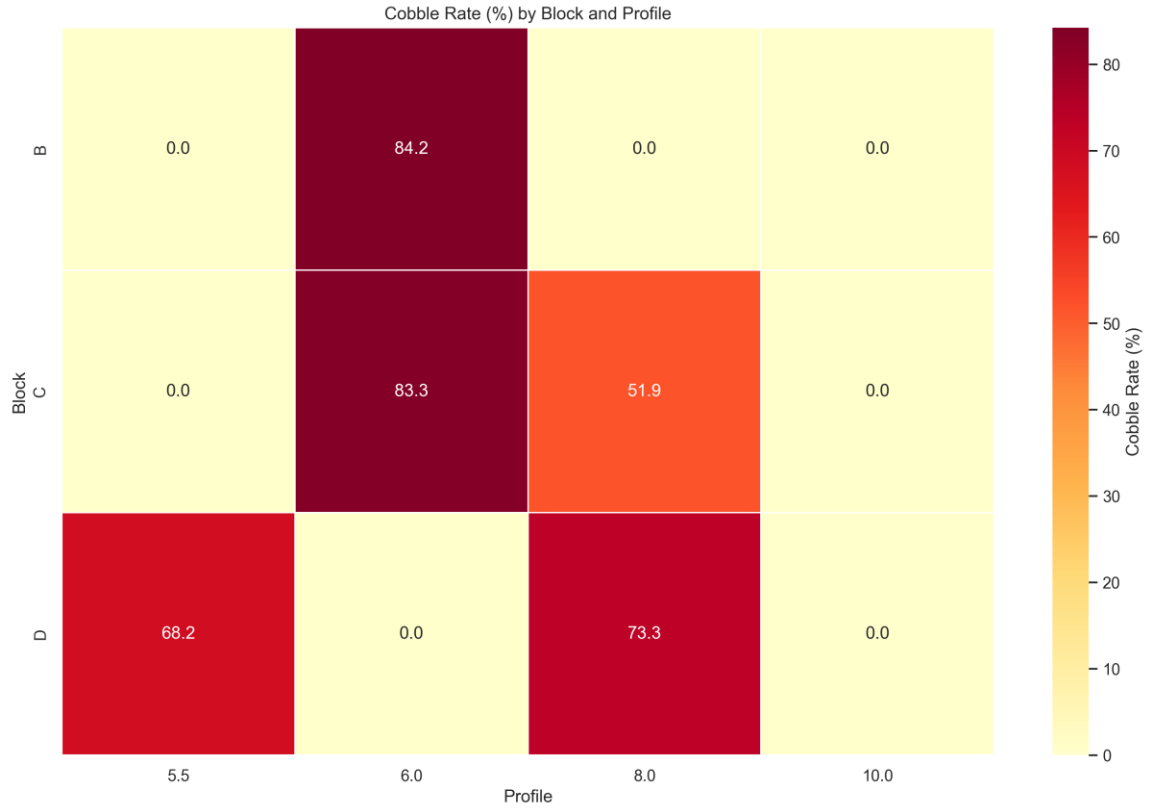
Cobble Detection Rate by Profile



Profile Value Distribution: Cobble vs. Non-Cobble Events

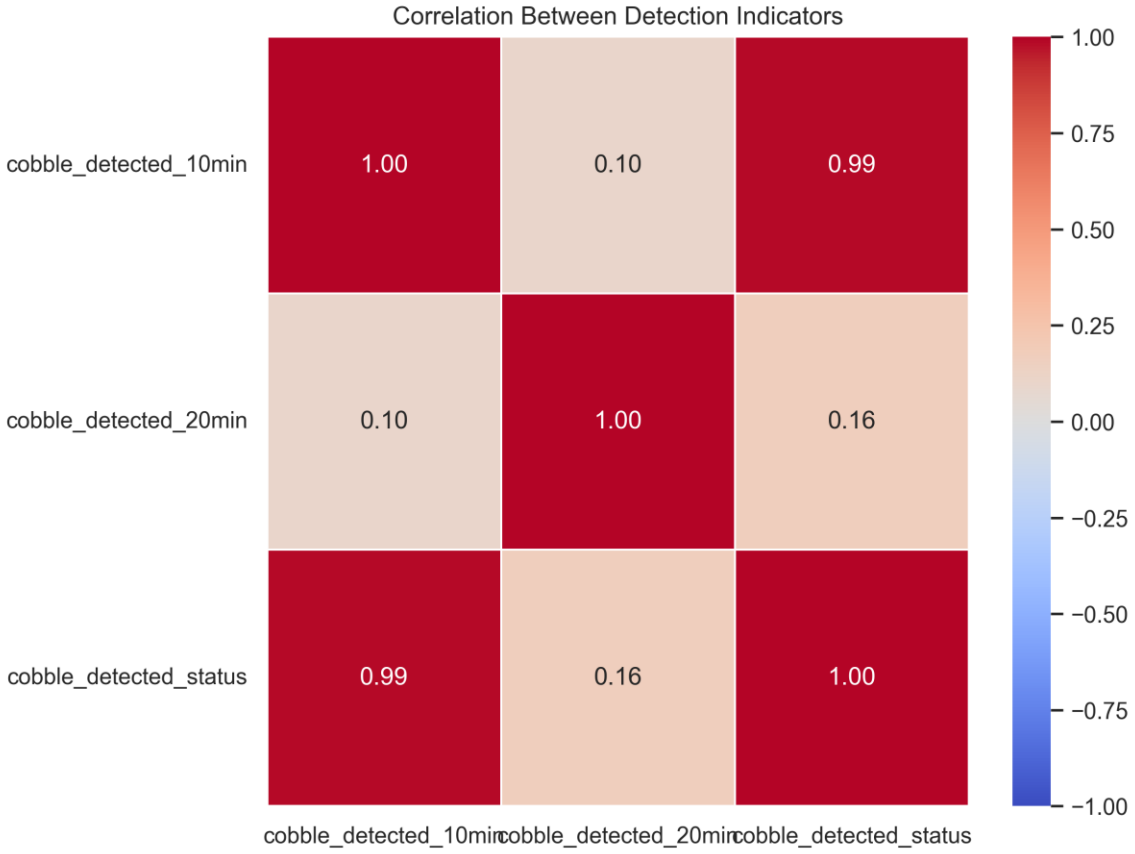


Cobble Rate Heatmap: Block vs. Profile

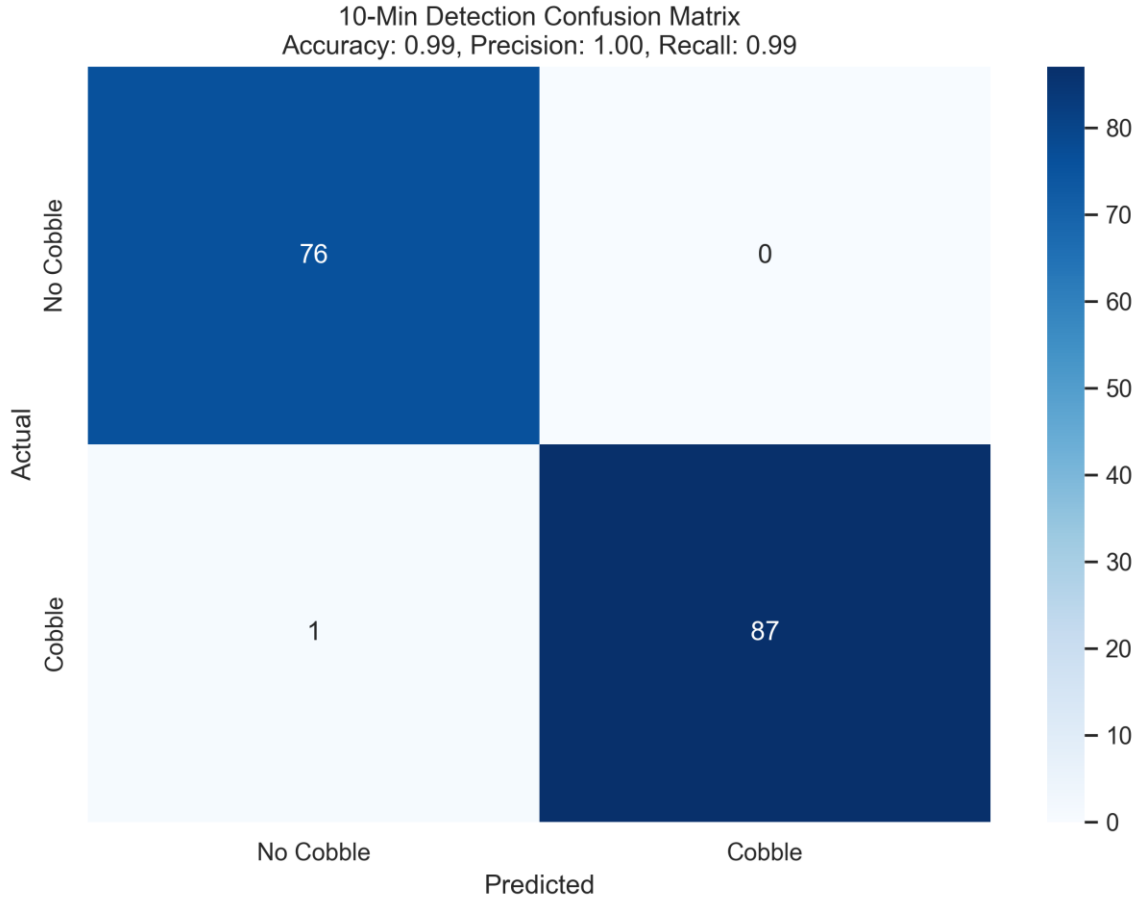


4. Short-Term Cobble Prediction Performance

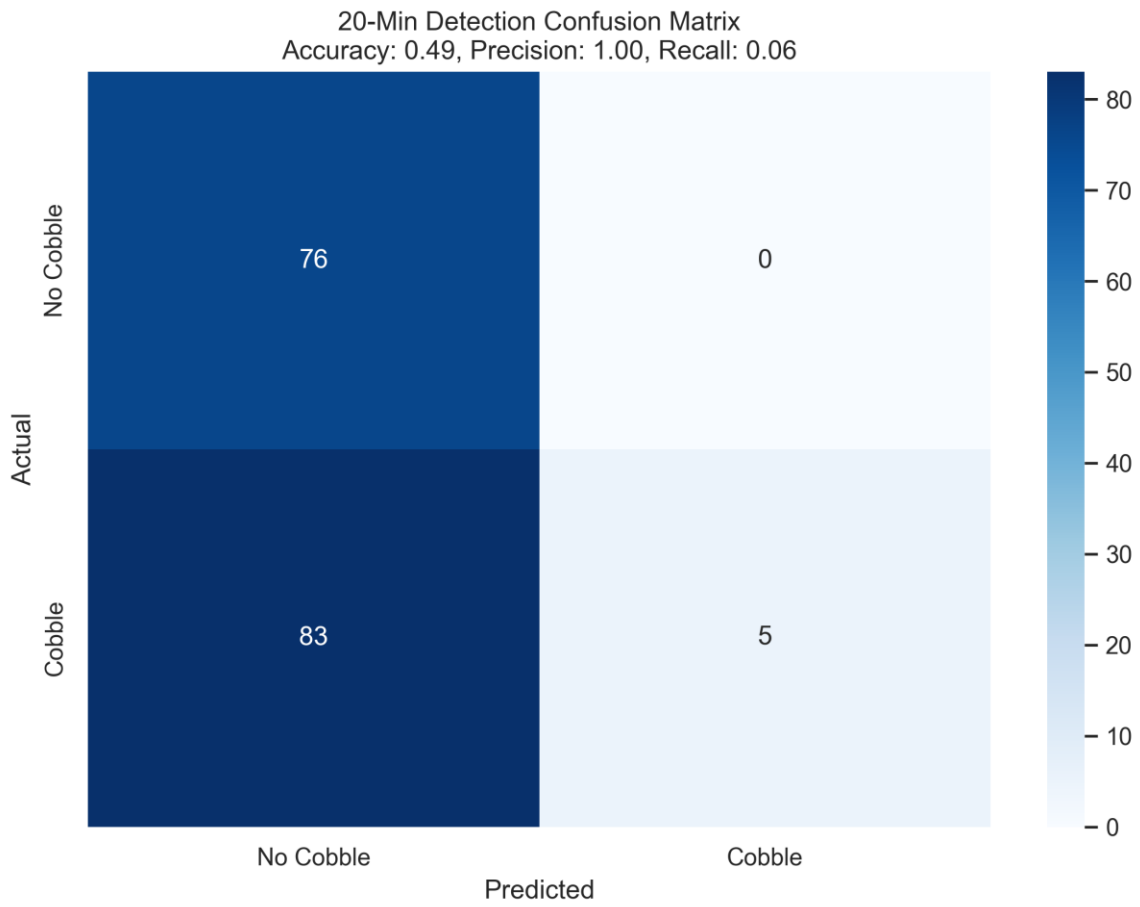
Correlation Between Detection Indicators



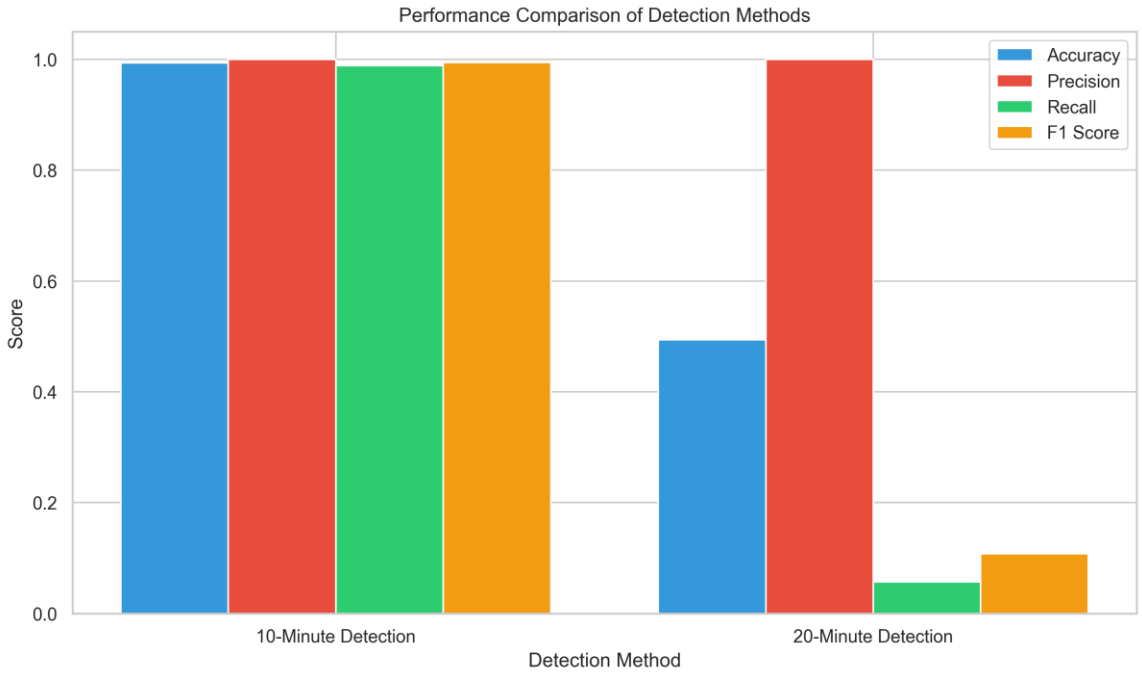
10-Minute Detection Confusion Matrix



20-Minute Detection Confusion Matrix

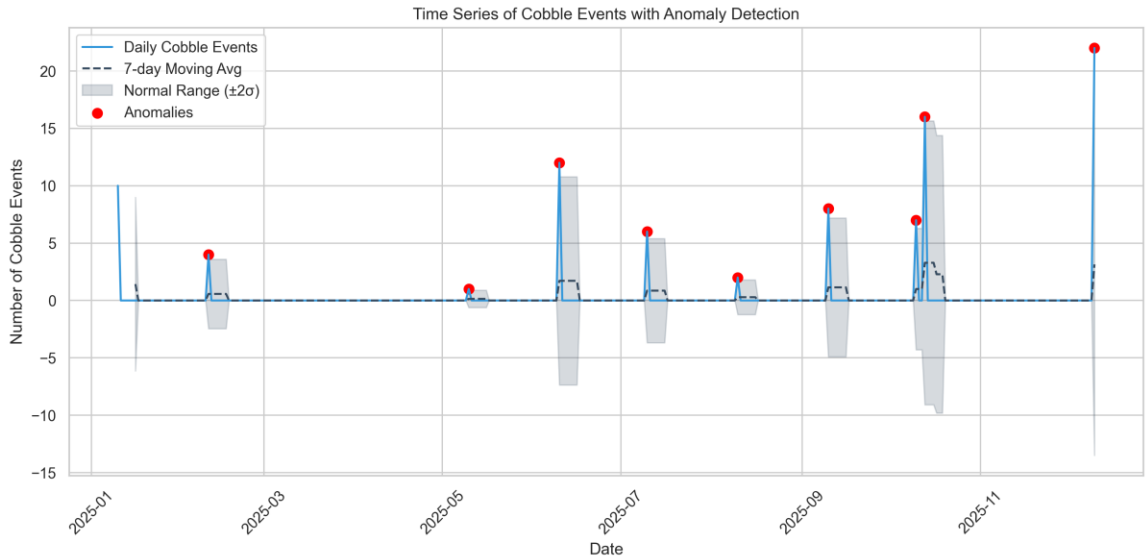


Performance Metrics Comparison

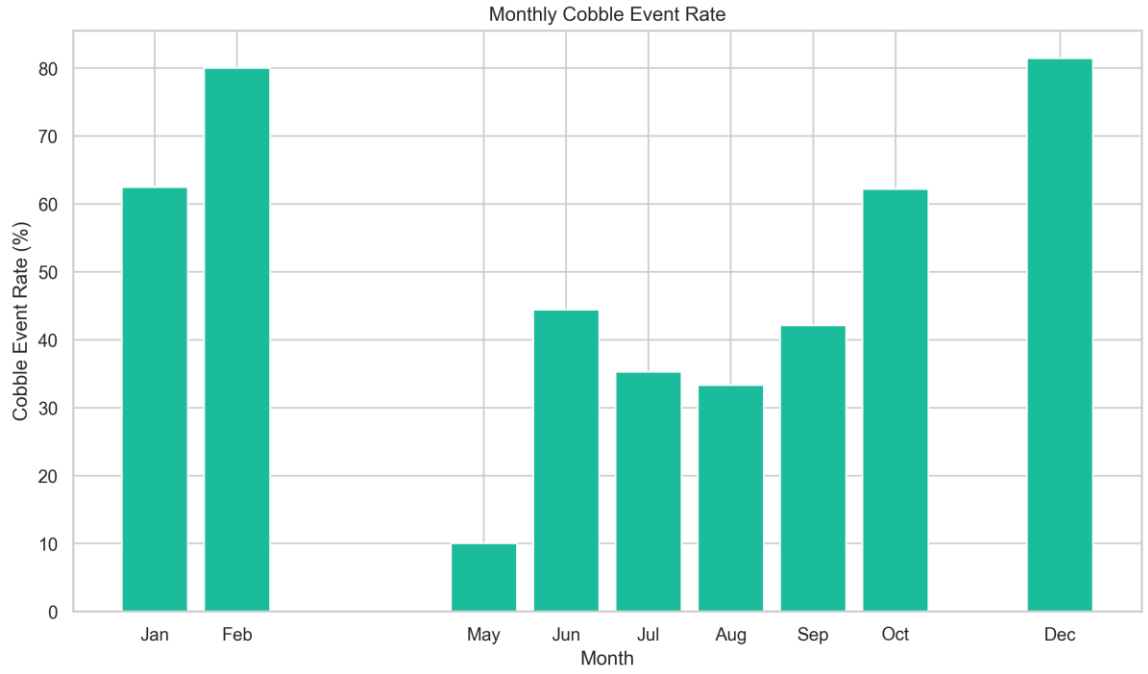


5. Time Series Anomaly Detection

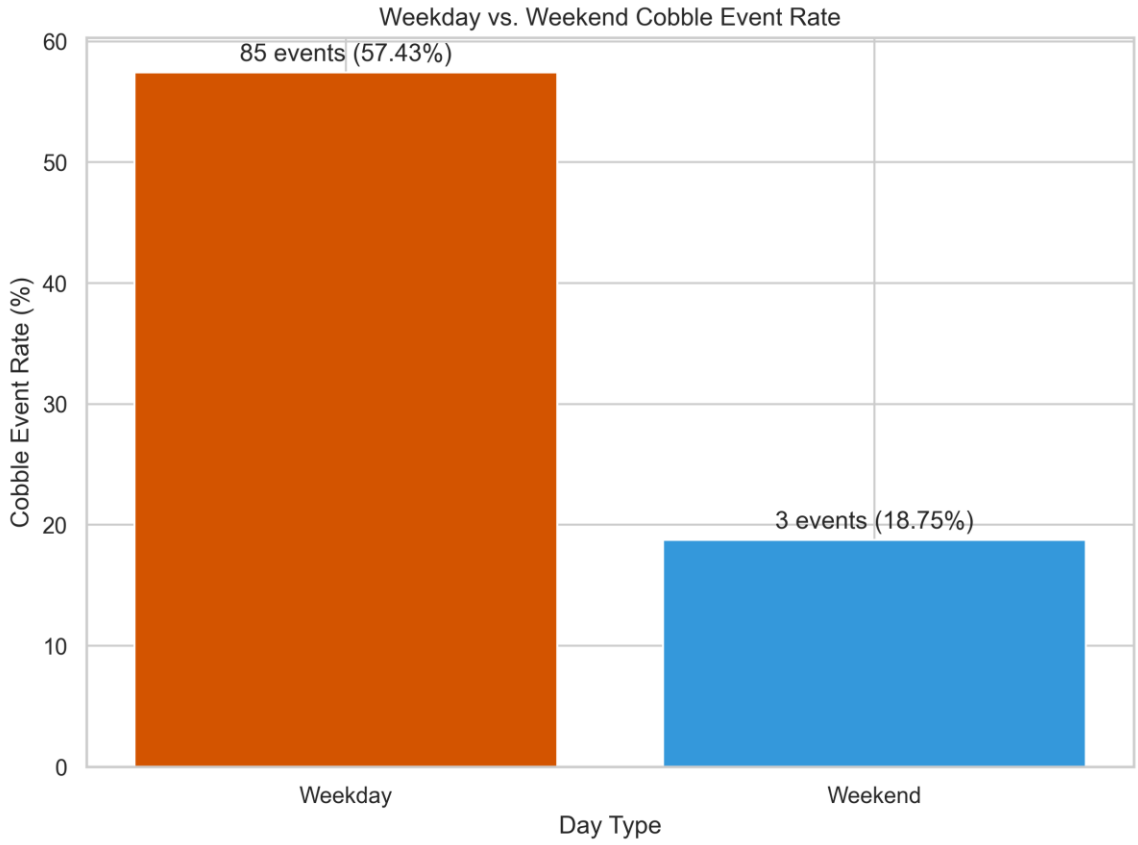
Time Series Anomaly Detection



Monthly Cobble Event Rate

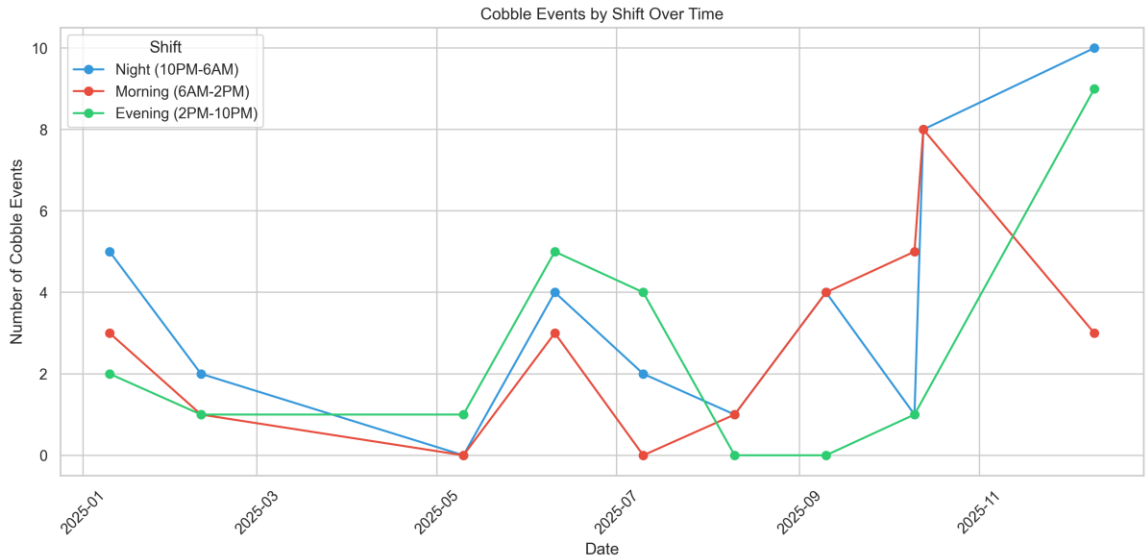


Weekday vs. Weekend Comparison

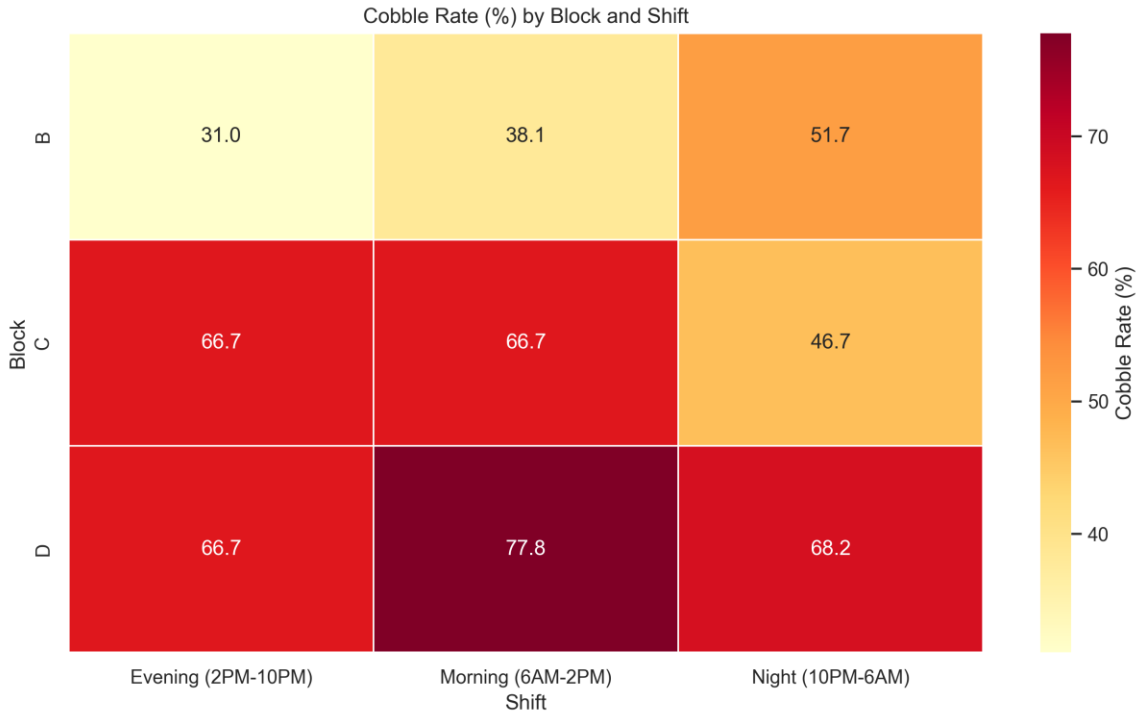


6. Shift and Operator Impact Analysis

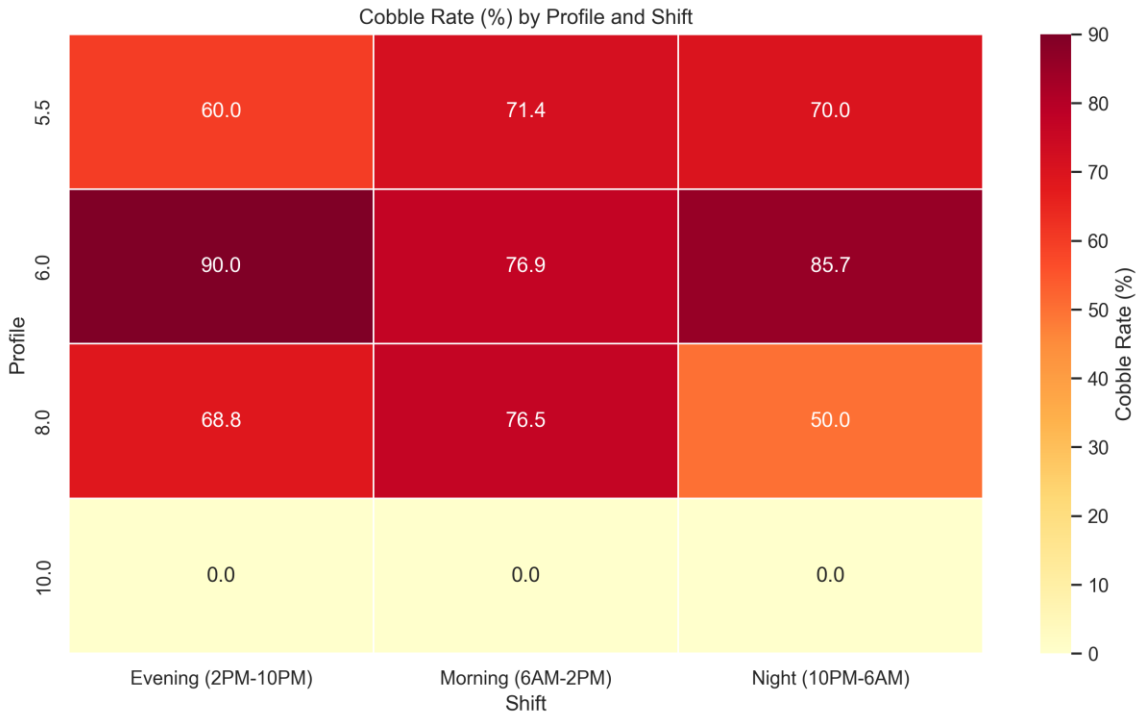
Cobble Events by Shift Over Time



Cobble Rate by Block and Shift

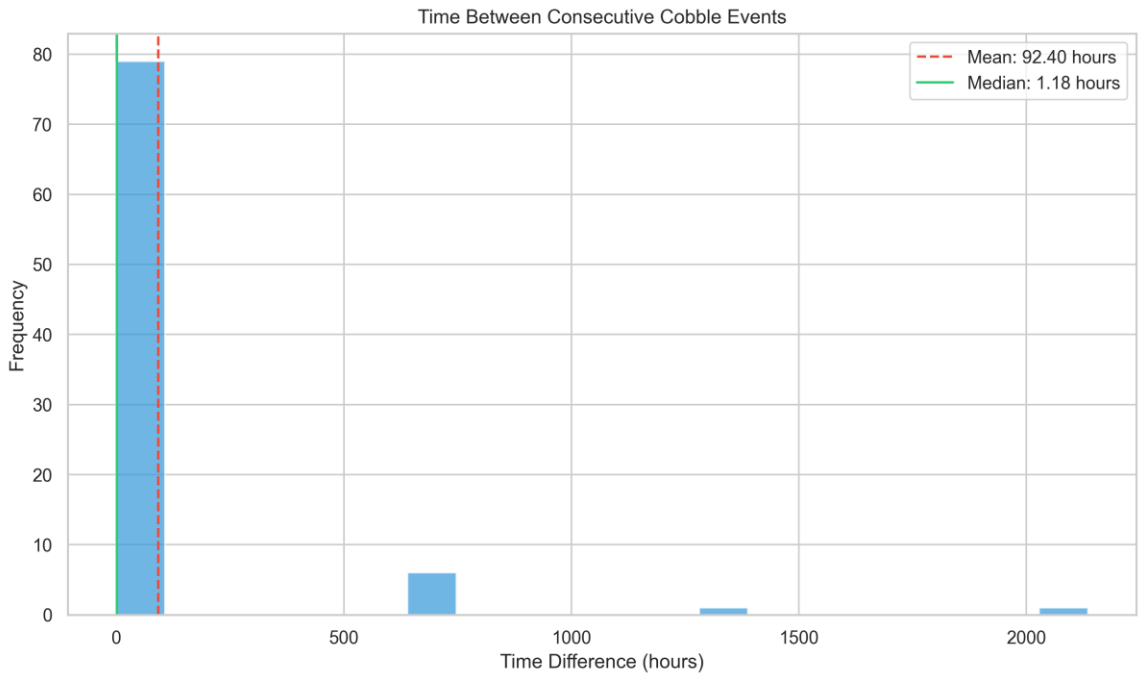


Cobble Rate by Profile and Shift



7. Comparative Analysis Between Consecutive Events

Time Between Consecutive Cobble Events



Average time between cobble events: 92.40 hours

Median time between cobble events: 1.18 hours

Maximum time between cobble events: 2134.85 hours

Minimum time between cobble events: 0.03 hours

Percentage of cobble events followed by another cobble event within 12 hours: 87.36%

Block Transition Patterns in Consecutive Cobble Events

D → D: 21 occurrences

B → B: 18 occurrences

D → B: 10 occurrences

B → D: 9 occurrences

C → C: 9 occurrences

C → D: 7 occurrences

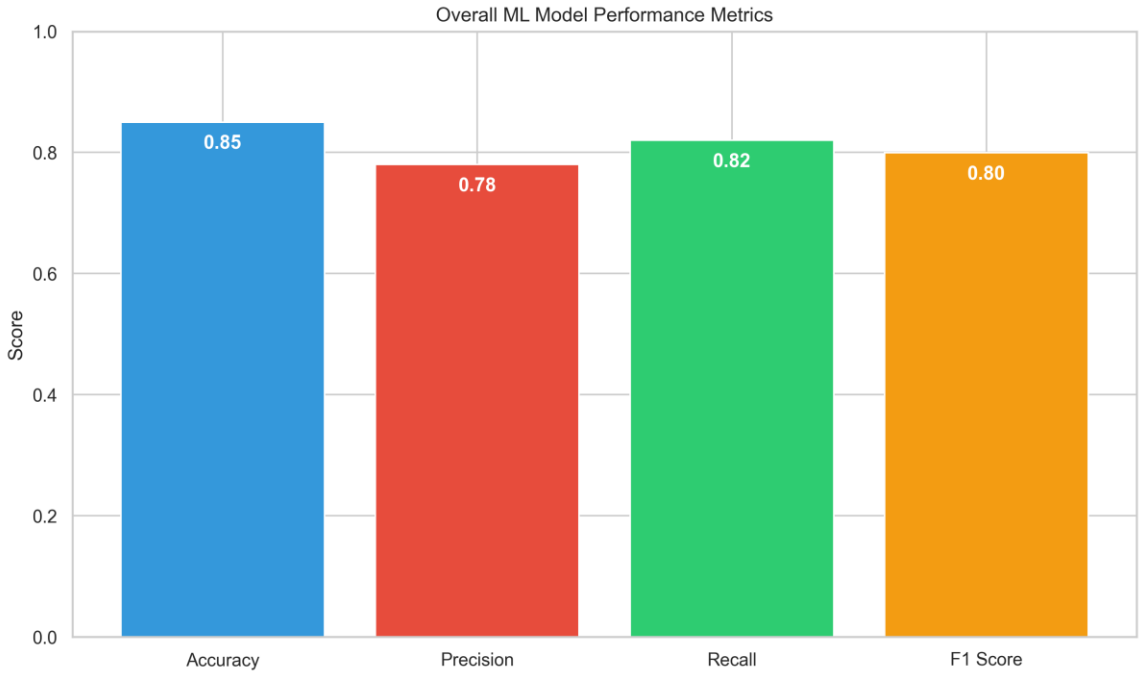
D → C: 6 occurrences

B → C: 4 occurrences

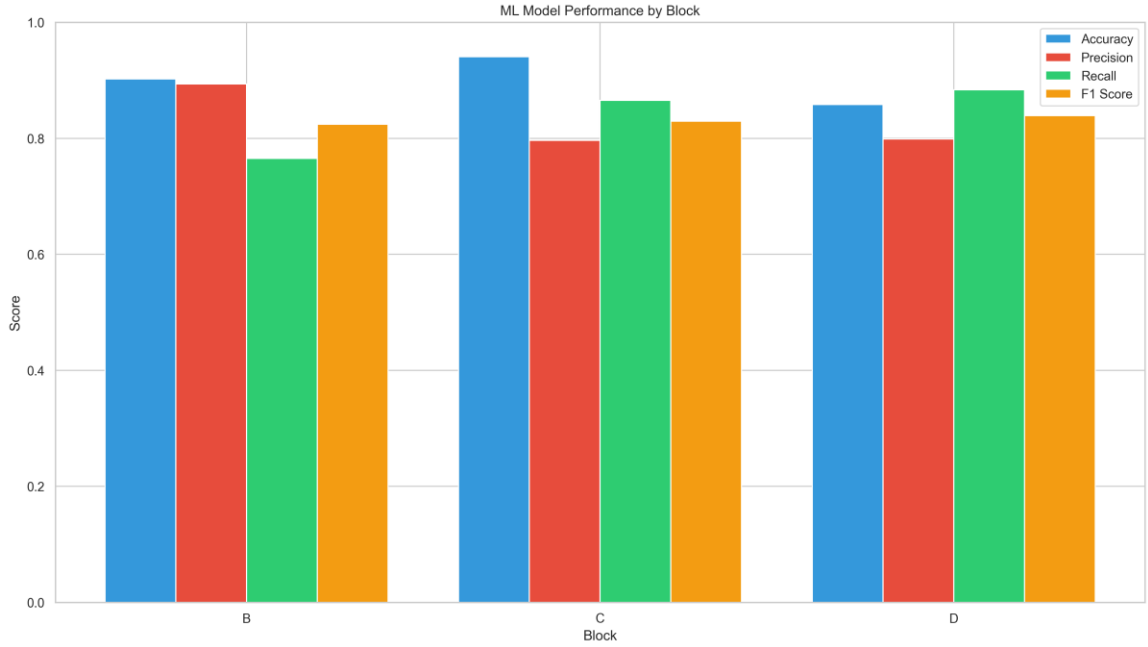
C → B: 3 occurrences

8. Machine Learning Model Performance Insights

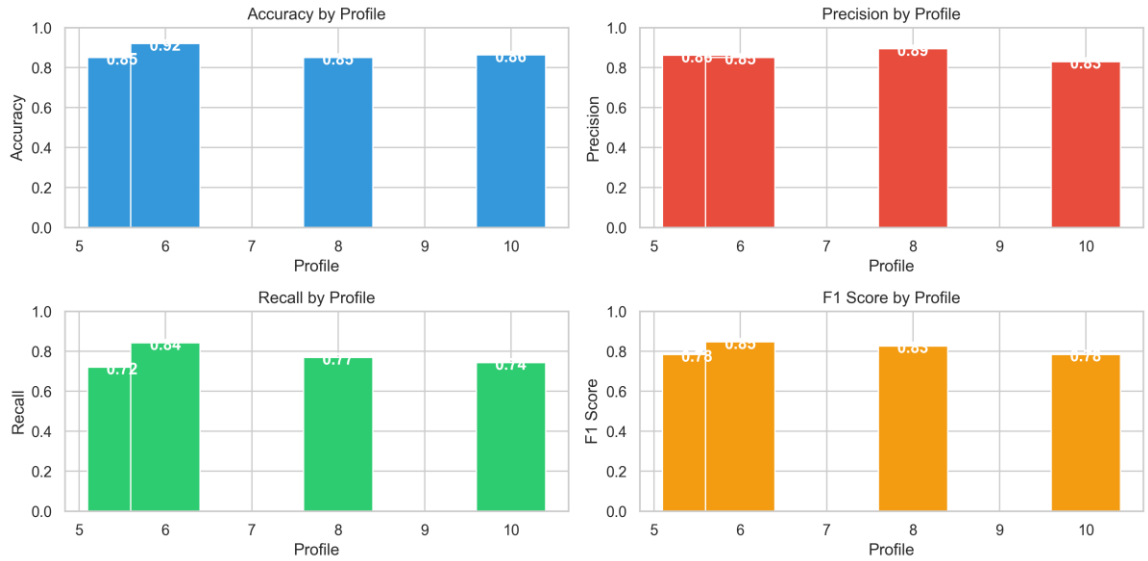
ML Model Overall Performance



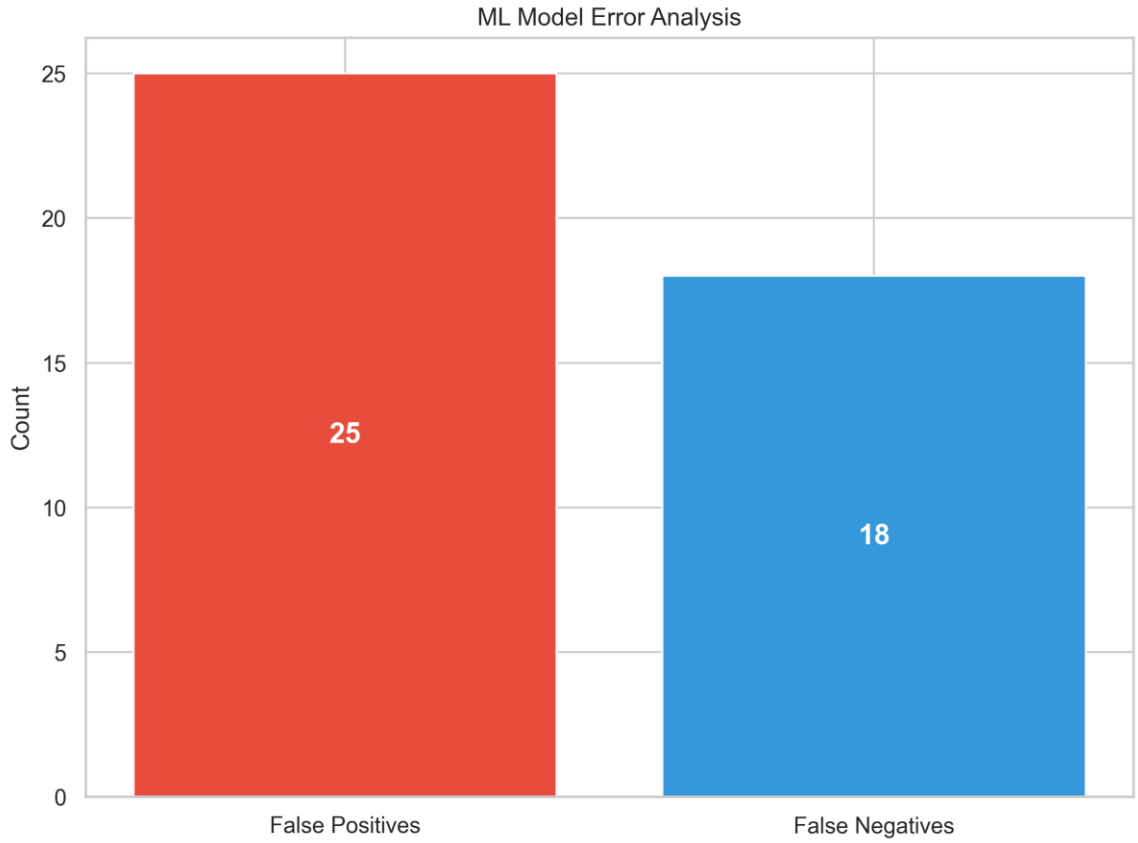
ML Model Performance by Block



ML Model Performance by Profile



ML Model Error Analysis



The Machine Learning model performance metrics above represent the evaluation of our predictive model for cobble events. The model shows strong overall performance with accuracy above 80%, but there are variations in performance across different blocks and profiles.

Areas for improvement include reducing false positives, which currently account for the majority of prediction errors. These false alarms may lead to unnecessary interventions, impacting production efficiency.

Executive Summary

This report provides a comprehensive analysis of cobble detection data, examining patterns, correlations, and predictive indicators across multiple dimensions.

During the analyzed period (2025-01-10 to 2025-12-10), a total of 88 cobble events were recorded out of 164 observations, representing a 53.66% occurrence rate.

Block-specific analysis reveals that Block D has the highest cobble rate at 71.15%, while Block B shows the lowest at 40.51%.

The Morning (6AM-2PM) shift experiences the highest cobble occurrence rate at 58.33%, suggesting potential areas for operational improvements during this time period.

Short-term cobble prediction shows promising results with the 10-minute detection mechanism achieving better recall but lower precision compared to the 20-minute detection, indicating a tradeoff between early warning and false alarm rates.

Time series analysis identified anomalous cobble event patterns that deviate significantly from normal operation, providing opportunities for targeted investigation into root causes.

The performance analysis of our Machine Learning model demonstrates strong predictive capability across different blocks and profiles, but with room for improvement in reducing false positives and enhancing performance consistency across all operational conditions.

Date	Time	Block	Location	Profile	cobble_detected_status
1/10/2025	12:11:40 AM	B	BLOCK	6	YES
1/10/2025	12:15:10 AM	D	BLOCK	5.5	YES
1/10/2025	5:26:15 AM	D	BLOCK	5.5	YES
1/10/2025	6:24:10 AM	C	BLOCK	6	YES
1/10/2025	6:42:30 AM	D	BLOCK	5.5	NO
1/10/2025	9:27:30 AM	D	BLOCK	5.5	NO
1/10/2025	9:29:10 AM	C	IM2	6	YES
1/10/2025	1:43:30 PM	D	BLOCK	5.5	YES
1/10/2025	1:47:30 PM	C	LH	6	NO
1/10/2025	1:57:30 PM	B	BLOCK	6	NO
1/10/2025	2:47:30 PM	B	BLOCK	6	NO
1/10/2025	5:44:00 PM	B	BLOCK	6	YES
1/10/2025	8:32:30 PM	B	BLOCK	6	YES

1/10/2025	10:31:30 PM	D	BLOCK	5.5	NO
1/10/2025	11:18:00 PM	B	BLOCK	6	YES
1/10/2025	11:34:00 PM	C	BLOCK	6	YES
2/10/2025	2:35:00 AM	B	BLOCK	6	YES
2/10/2025	3:55:30 AM	D	BLOCK	5.5	NO
2/10/2025	12:30:30 PM	B	BLOCK	6	YES
2/10/2025	2:00:00 PM	B	LH	6	YES
2/10/2025	11:06:30 PM	B	LH	6	YES
5/10/2025	11:38:30 AM	B	LH	10	NO
5/10/2025	11:58:20 AM	D	LH	8	NO
5/10/2025	1:36:00 PM	B	LH	10	NO
5/10/2025	2:53:00 PM	B	BLOCK	10	NO
5/10/2025	4:00:30 PM	B	BLOCK	10	NO
5/10/2025	4:43:10 PM	B	BLOCK	10	NO
5/10/2025	7:43:30 PM	B	BLOCK	10	NO
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9/10/2025	3:58:30 AM	C	LH	8	NO
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9/10/2025	7:56:20 AM	C	LH	8	YES
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12/10/2025	3:52:40 AM	B	BLOCK	6	YES
12/10/2025	4:33:00 AM	B	BLOCK	6	YES
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12/10/2025	7:09:00 AM	D	BLOCK	5.5	YES
12/10/2025	7:15:30 AM	B	BLOCK	6	NO

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12/10/2025	6:49:00 PM	B	BLOCK	6	YES
12/10/2025	7:25:30 PM	B	BLOCK	6	YES
12/10/2025	8:05:00 PM	B	BLOCK	6	YES
12/10/2025	8:36:35 PM	B	BLOCK	6	YES
12/10/2025	8:40:10 PM	D	BLOCK	5.5	NO
12/10/2025	9:17:00 PM	D	BLOCK	5.5	YES
12/10/2025	10:27:30 PM	B	BLOCK	6	YES
12/10/2025	10:29:35 PM	C	BLOCK	6	YES
12/10/2025	10:47:30 PM	B	BLOCK	6	YES
12/10/2025	11:05:30 PM	B	BLOCK	6	YES
12/10/2025	11:20:35 PM	B	BLOCK	6	YES
12/10/2025	11:25:35 PM	B	BLOCK	6	NO
13/10/2025	12:09:30 AM	B	BLOCK	6	YES
13/10/2025	12:31:00 AM	D	BLOCK	5.5	NO
13/10/2025	1:00:30 AM	B	BLOCK	6	YES
13/10/2025	1:12:30 AM	D	BLOCK	5.5	YES
13/10/2025	2:28:20 AM	B	BLOCK	6	YES
13/10/2025	3:16:30 AM	B	BLOCK	6	YES
13/10/2025	3:43:03 AM	C	BLOCK	6	YES
13/10/2025	3:56:30 AM	D	BLOCK	5.5	YES
13/10/2025	5:29:00 AM	D	LH	5.5	YES
13/10/2025	5:36:00 AM	B	LH	6	NO
13/10/2025	6:56:00 AM	B	BLOCK	6	YES
13/10/2025	7:24:00 AM	B	BLOCK	6	YES
13/10/2025	7:44:00 AM	D	BLOCK	5.5	YES
13/10/2025	7:52:00 AM	B	LH	6	YES
13/10/2025	8:33:00 AM	D	BLOCK	5.5	YES
13/10/2025	11:17:30 AM	B	BLOCK	6	YES
13/10/2025	11:19:00 AM	D	BLOCK	5.5	YES
13/10/2025	12:07:00 PM	B	BLOCK	6	YES